

HYPERTENSION



150
98
mm Hg

Hypertension, or high blood pressure, occurs when the force of the blood against the artery walls is too high.

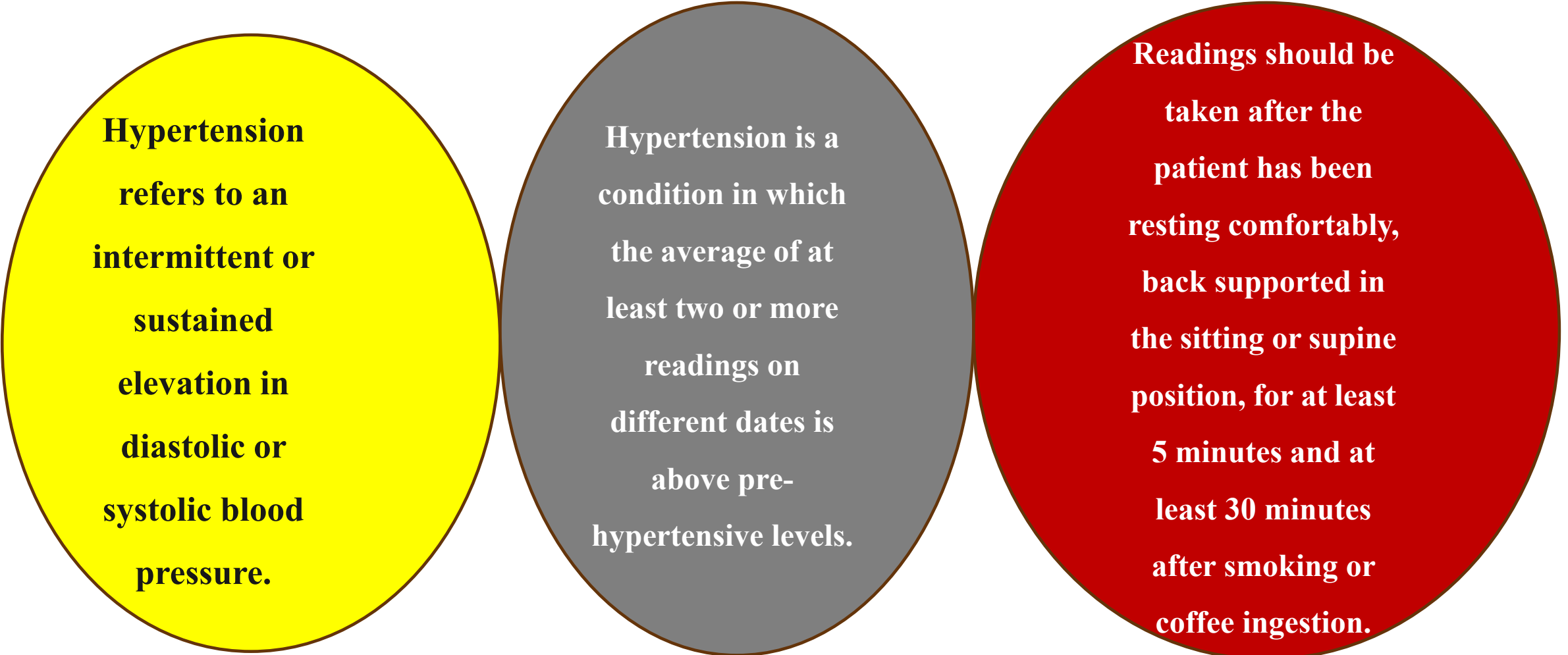


HYPERTENSION

AGENDA

- Definition
- Hemodynamic factors affecting blood pressure
- Types
- Causes
- Pathophysiology
- Clinical manifestation
- Diagnostic tests
- Complications
- Medical management
- Nursing management





**Hypertension
refers to an
intermittent or
sustained
elevation in
diastolic or
systolic blood
pressure.**

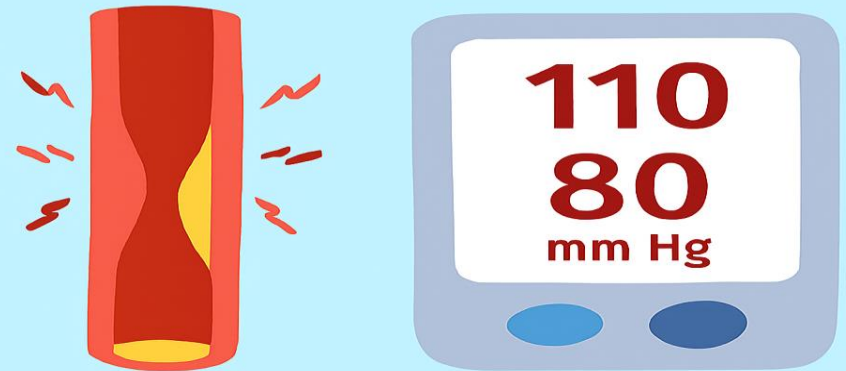
**Hypertension is a
condition in which
the average of at
least two or more
readings on
different dates is
above pre-
hypertensive levels.**

**Readings should be
taken after the
patient has been
resting comfortably,
back supported in
the sitting or supine
position, for at least
5 minutes and at
least 30 minutes
after smoking or
coffee ingestion.**



Arterial blood pressure is the force that the blood exerts against the walls of the aorta and its branches. $BP = COP \times PR$

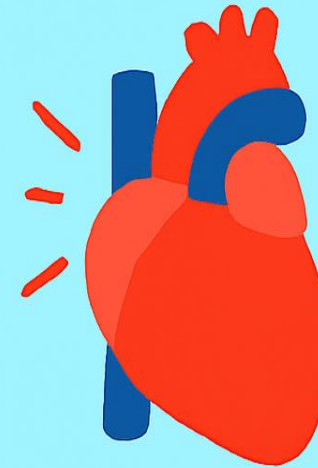
ARTERIAL BLOOD PRESSURE



Arterial blood pressure is the measurement of the force the blood exerts against the artery walls as it pumps blood throughout the body.

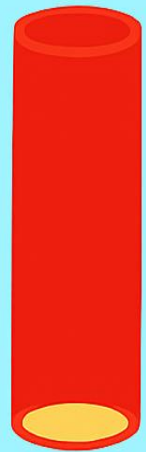
- The blood pressure is greatest during ventricular contraction, or **systole**, when blood is ejected into the aorta.
- Diastolic pressure is the pressure when the ventricles are in the relaxation phase, or **diastole**, just before the next contraction of the ventricles.

SYSTOLE

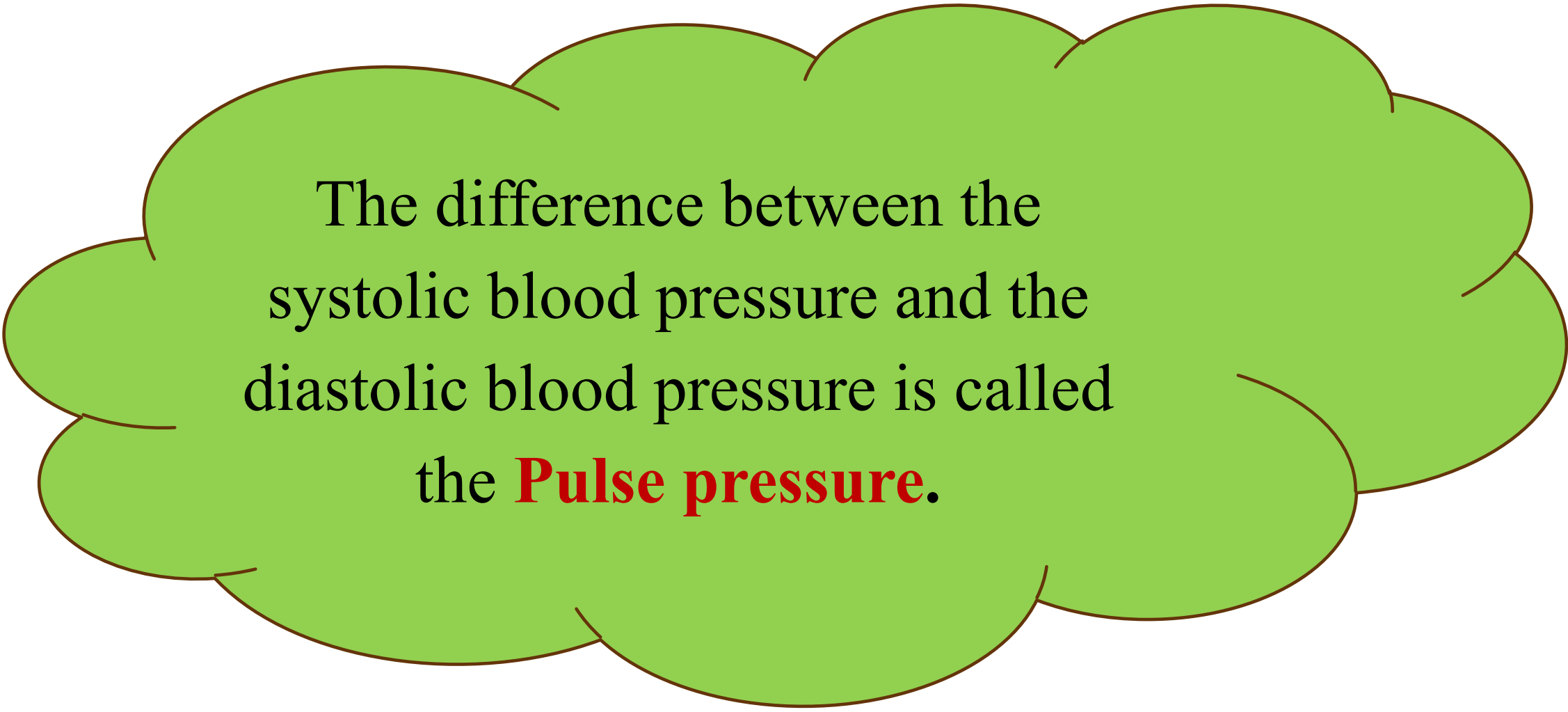


Heart contracts
and pushes blood
out of the heart

DIASTOLE



Heart relaxes
and fills with
blood



The difference between the
systolic blood pressure and the
diastolic blood pressure is called
the **Pulse pressure**.

CLASSIFICATION OF BLOOD PRESSURE IN ADULT

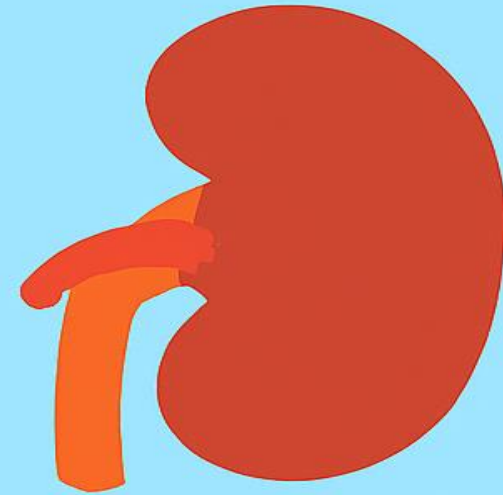
BP Classifications	Systolic BP	Diastolic BP
Normal	<120 mmHg	<80 mmHg
Prehypertension	120–139 mmHg	80–89 mmHg
Stage 1 hypertension	140–159 mmHg	90–99 mmHg
Stage 2 hypertension	≥ 160 mmHg	≥ 100 mmHg

ESSENTIAL OR PRIMARY HYPERTENSION



Chronic elevation of blood pressure from unknown cause (idiopathic).

SECONDARY HYPERTENSION



Elevated blood pressure with an identifiable cause that is often correctable.

MALIGNANT HYPERTENSION

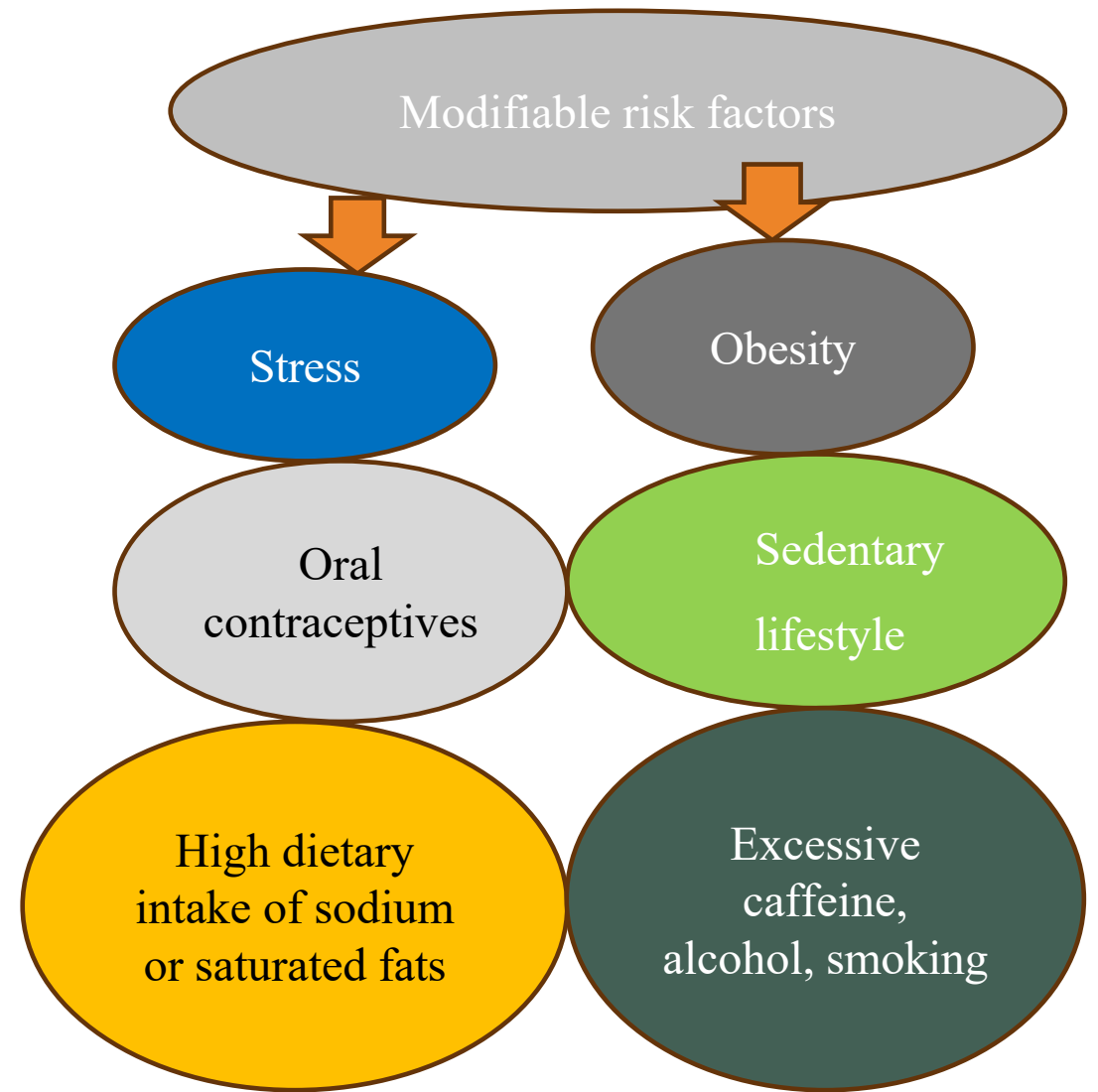
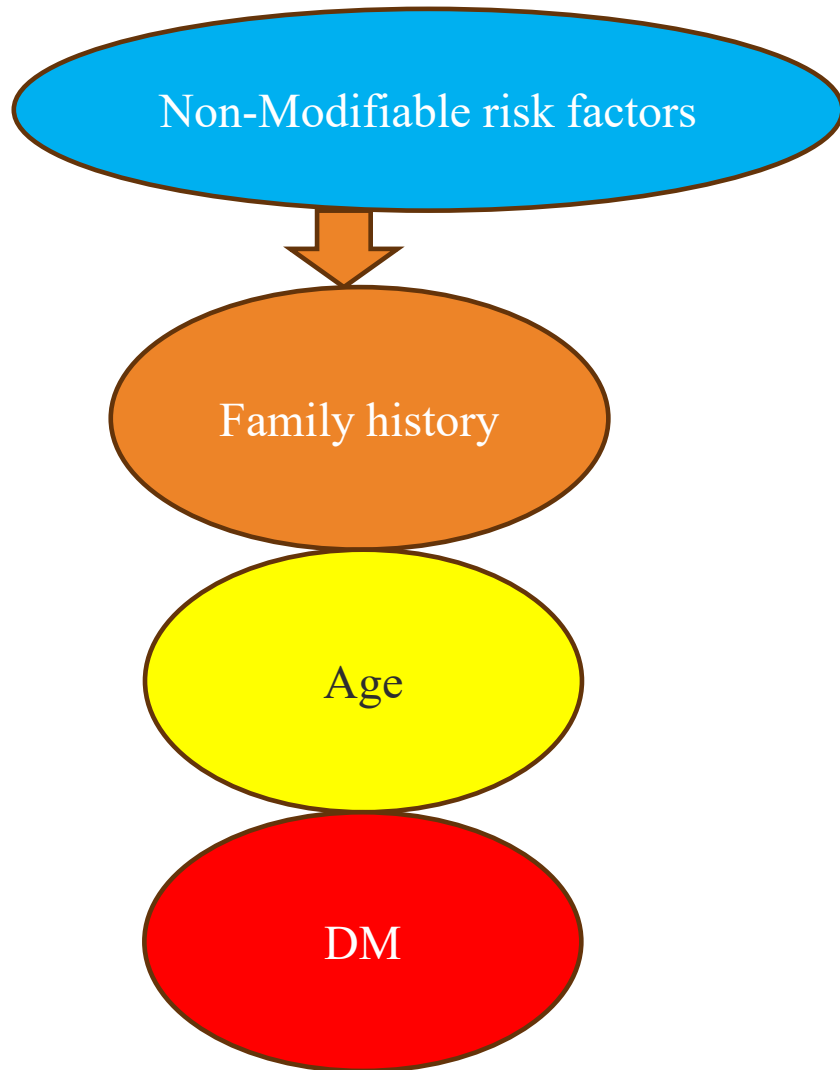


**Severe, fulminant form
of hypertension common
to both types**

ISOLATED SYSTOLIC HYPERTENSION



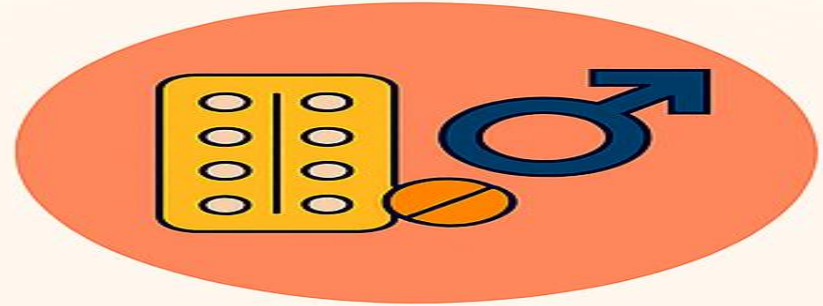
**Systolic pressure of 140 mm Hg
or greater and a diastolic
pressure of 90 mm Hg or less.**



Causes of secondary hypertension



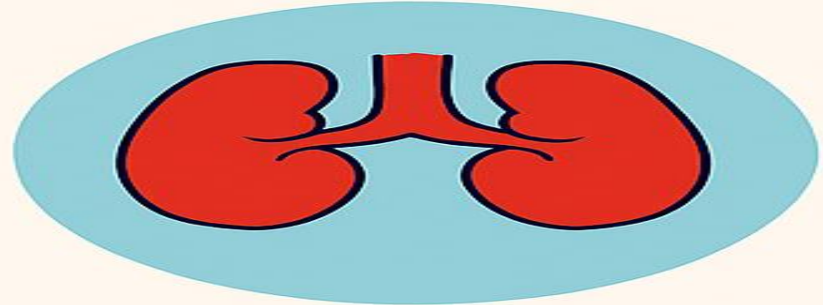
Sleep apnea



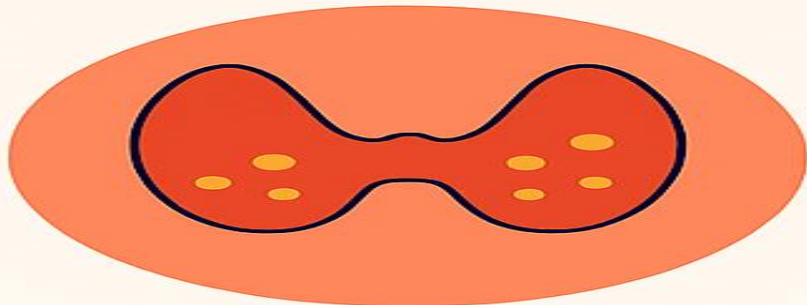
Estrogen use



**Long-term
corticosteroid therapy**



**Thyroid or parathy-
roid disease**



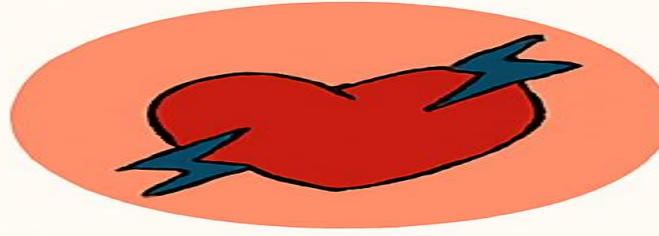
❑ **Systolic blood pressure** is the pressure that the heart pumps against to force blood from the left side of the heart to the aorta and to major arteries.

❑ **Diastolic blood pressure** is the pressure required to permit filling of the ventricles before the next systole cycle.

Signs and symptoms of hypertension



Headache



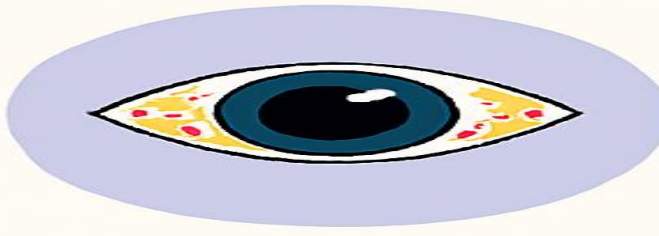
**Irregular
heartbeat**



Chest pain



Tinnitus



**Visual
changes**



Fatigue

**Signs and symptoms
of hypertension**

COMPLICATIONS

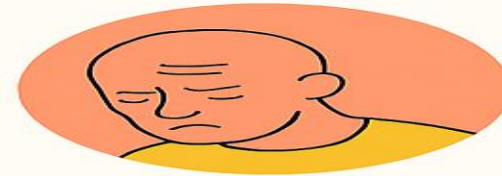
Complications of hypertension



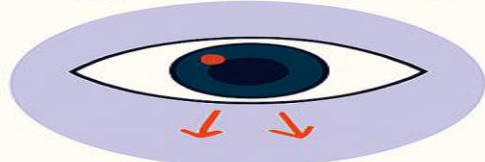
Left ventricular hypertrophy



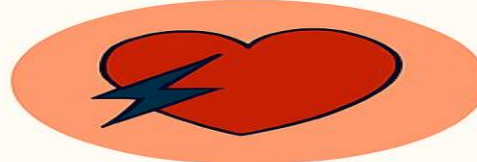
Angina



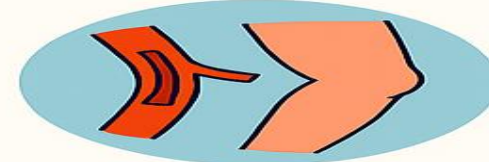
Angina



Sexual dysfunction



MI – Myocardial Infarction



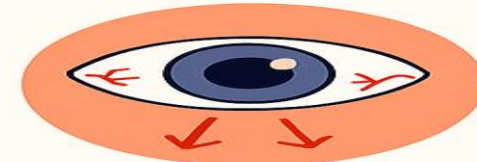
Peripheral arterial disease



Transient ischemic attack



Nephropathy



Retinopathy

INVESTIGATIONS

Measures Blood
pressure

Chest x-ray

Monitor BUN
blood urea
nitrogen

Check
electrolytes for
imbalance

Ophthalmologic
test

Glucose test

TREATMENT

- Lifestyle changes
- Reduce caloric intake and exercise

Drug	Mechanism of Action	Side Effects	Nursing Considerations
Beta Adrenergic Blockers propranolol (Inderal)	Decrease renin levels	Hypotension, bradycardia, CHF, depression, hypoglycemia	• Administer with meals. • Advise clients not discontinue medication abruptly. • Monitor for fluid retention, and difficulty breathing. • Monitor blood glucose level in clients with diabetes.
Alpha- Adrenergic Blockers prazosin	Relax vascular smooth muscle	Hypotension, dizziness, drowsiness, nausea	• Administer first dose just before bedtime to reduce potential for syncope. • Monitor client for postural hypotension. • Caution client to change positions slowly.

NURSING INTERVENTIONS

- ✓ Monitor blood pressure with multiple readings—lying, sitting, and standing, bilateral both arms.
- ✓ Record fluid intake and output.
- ✓ Reduce stress by providing a quiet environment.

EXPLAIN TO THE PATIENT:

No smoking

Change to a
low-sodium

Lower
cholesterol
intake

Reduce
weight

Reduce
alcohol intake

Exercise
regularly



Adhere to the
treatment
regimen

Avoid hot
baths

SAFETY MEASURES TO PREVENT FALLS FOR PATIENTS WITH ORTHOSTATIC HYPOTENSION

Rise slowly from a lying to a sitting position

When walking, do not turn the head or body abruptly

Report sudden dizziness

When feeling unsteady while standing, call for assistance before walking.

Use the bathroom before meals and try to avoid getting up for 30 to 60 minutes after meals

Thank
You